

Jayasurya Bhaskar

[PORTFOLIO LINK](#)

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SUMMARY

AI Engineer with **3+ years** building end-to-end AI software across **computer vision**, **vision-language models**, and **LLM applications**. Builds **scalable AI pipelines** from data ingestion to production deployment, with company-wide ownership of **MLOps**. Comfortable in **fast-paced R&D**, from **MVP prototype to production**.

AREAS OF SPECIALIZATION

• Computer Vision • Vision-Language Models • Deep Learning • MLOps and Model Lifecycle • Production AI Pipelines • Cloud Deployment • Full-Stack AI Applications

EXPERIENCE

AI Engineer

May 2024 – Present

Castillo Engineering, Orlando, FL

- Built **Castillo Planset QC**, a production VLM pipeline on **OpenAI** and **Gemini Vision** reviewing solar and electrical drawings against **470+ compliance checks** and **18 deterministic NEC rules**, cutting a **5-hour** plan review to **minutes**.
- Engineered adaptive tiling and coordinate remapping so **Gemini Vision** reads large-format drawings at native resolution and returns each finding to its exact sheet coordinate.
- Built **PMO 360** on **FastAPI**, **PostgreSQL**, **React**, and the **OpenAI API**, turning meeting notes into tracked actions and client deliverables across every PM team.
- Trained a **MobileNetV3** classifier with **OpenCV** preprocessing to route solar-installation photos to **Gemini Vision**, covering a **350+ report/month** permit workflow.
- Built **PySpark** pipelines that standardize roughly **3 TB** of raw solar-plan data into **Delta Lake** for downstream CV and analytics.
- **Own the company-wide codebase** and release workflow: PR reviews, **GitHub** administration, **CI/CD**, and mentoring interns.

Machine Learning Engineer

Feb 2022 – Aug 2022

Genpact, Remote

- Built a **YOLOv5 + U-Net** pipeline in **PyTorch** and **OpenCV** that localizes teeth and segments periodontal bone on panoramic dental X-rays, deployed across **5 dental clinics**.
- Trained a **GAN** model that cleaned noisy, low-contrast radiographs for reliable downstream detection.
- Built vehicle-damage pipelines over **100K+ images**, automating dataset versioning with **PySpark** and **Airflow** and promoting models through **MLflow** and **CI/CD**.

PROJECTS

BLIP: Burst Localisation & Image Pruning

[Live app](#)

Groups near-duplicate photos with **DINOv3** embeddings, ranks the best frame, and drives **context search with LLMs**.

Engineering Planset QC Automation

[Demo](#) | [Code](#)

VLM plan-set **compliance** review with **tiled inference** and coordinate-accurate findings.

PMO 360

[Demo](#) | [Code](#)

All-in-One **Project Management portal** for creating meeting minutes, action items and tracking tasks using LLMs and APIs.

EDUCATION

M.S. Computer Science, University of Central Florida, 2024

GPA: **3.7/4.0**

Graduate Teaching Assistant. Advanced AI: Neural Networks, Deep Learning, NLP, Reinforcement Learning.